

Anatomy foundations

Know the direction. Name the plane.

A focused set of references and original exercises for learning the language of location.

28

practice questions

10

directional terms

8

printable pages

Use the pack in three passes

1. Read the two reference pages before attempting the exercises.
2. Complete pages 4-6 from memory, using anatomical position for every comparison.
3. Check pages 7-8. Explain why each missed answer fits, then retry that item.

Inside

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Print on US Letter paper at 100% scale. Work directly on these pages or use a separate notebook.

Five pairs, ten terms

Direction is a comparison: name the structure you are describing and the structure you are comparing it with.

Set the reference position. Picture a person standing upright, facing forward, arms at the sides and palms facing forward. Right and left belong to that person.

Superior

Higher, toward the head.
The head is superior to the chest.

Inferior

Lower, toward the feet.
The ankle is inferior to the knee.

Anterior

Toward the front.
The sternum is anterior to the heart.

Posterior

Toward the back.
The vertebral column is posterior to the sternum.

Medial

Nearer the body's midline.
The nose is medial to the ears.

Lateral

Farther from the body's midline.
The ears are lateral to the nose.

Proximal

Nearer a limb's attachment to the trunk.
The elbow is proximal to the wrist.

Distal

Farther from a limb's attachment to the trunk.
The fingers are distal to the elbow.

Superficial

Nearer the body surface.
The skin is superficial to skeletal muscle.

Deep

Farther beneath the body surface.
The brain is deep to the skull.

Use the limb's attachment for proximal/distal and the body surface for superficial/deep.

The three main planes

A plane is an imaginary flat surface through a three-dimensional body. Learn the portions it separates.

Sagittal

Right and left portions.

Conceptual portions; diagrams are not body views.



Frontal (coronal)

Anterior and posterior portions.

Conceptual portions; diagrams are not body views.



Transverse (axial)

Superior and inferior portions.

Conceptual portions; diagrams are not body views.



Sagittal does not always mean equal halves. A midsagittal (median) plane is at the midline and creates equal right and left halves. A parasagittal plane is offset from that midline.

A vertical plane can be sagittal or frontal. Ask what it separates, rather than relying on the word vertical.

Name the relationship

Write the most appropriate directional term. Each sentence uses standard anatomical position.

01 The head is _____ to the chest.

02 The ankle is _____ to the knee.

03 The sternum is _____ to the heart.

04 The vertebral column is _____ to the sternum.

05 The nose is _____ to the ears.

06 The ears are _____ to the nose.

07 The elbow is _____ to the wrist.

08 The fingers are _____ to the elbow.

09 The skin is _____ to the underlying skeletal muscle.

10 The brain is _____ to the skull.

11 The shoulder is _____ to the elbow.

12 The wrist is _____ to the shoulder.

Use the comparison

Write one precise answer. For a reversed comparison, choose the opposite directional term.

- 13** A marker moves along an upper limb from the wrist toward the elbow. Which direction does it travel?
-
- 14** A marker moves along a lower limb from the knee toward the ankle. Which direction does it travel?
-
- 15** A marker moves from the side of the chest toward the breastbone. Which direction does it travel?
-
- 16** A marker moves from the centre of the chest toward the side. Which direction does it travel?
-
- 17** The brain is deep to the skull. Complete the reversed comparison: the skull is _____ to the brain.
-
- 18** The sternum is anterior to the heart. Complete the reversed comparison: the heart is _____ to the sternum.
-
- 19** A person faces you. Their right hand appears on which side of your view?
-
- 20** A person lies face up. What term describes that position?
-

Planes and positions

Use the plane or position name. Accepted synonyms appear in the answer key.

21 Which plane divides the body into right and left portions?

22 Which plane divides the body into anterior and posterior portions?

23 Which plane divides the body into superior and inferior portions?

24 Which specific sagittal plane passes through the midline to produce equal right and left halves?

25 A sagittal plane is offset from the midline and produces unequal right and left portions. What is its more specific name?

26 In standard anatomical position, do the palms face forward or backward?

27 Which body-position term means lying face down?

28 A person changes from standing to lying down. Do anatomical right and left switch?

Answers: 01-16

Check the term and its reason. A correct answer should describe the relationship, not just match a memorised phrase.

- 01 Superior.** The head lies above the chest.
- 02 Inferior.** The ankle lies below the knee.
- 03 Anterior.** The sternum is in front of the heart.
- 04 Posterior.** The vertebral column lies behind the sternum.
- 05 Medial.** The nose is closer to the body's midline.
- 06 Lateral.** The ears lie farther from the body's midline.
- 07 Proximal.** The elbow is nearer the upper limb's attachment to the trunk.
- 08 Distal.** The fingers lie farther along the limb from the trunk.
- 09 Superficial.** Skin lies nearer the outside surface.
- 10 Deep.** The brain lies farther beneath the body surface.
- 11 Proximal.** The shoulder is nearer the upper limb's attachment.
- 12 Distal.** The wrist lies farther from the upper limb's attachment.
- 13 Proximal.** The marker moves toward the attachment of the limb to the trunk.
- 14 Distal.** The marker moves away from the attachment of the limb to the trunk.
- 15 Medial.** The marker moves toward the midline.
- 16 Lateral.** The marker moves away from the midline.

Answers: 17-28

- 17 Superficial.** Reversing the comparison reverses the directional term.
- 18 Posterior.** The heart lies behind the sternum.
- 19 Your left.** Anatomical right and left belong to the person being described.
- 20 Supine.** Supine means lying face up; prone means lying face down.
- 21 Sagittal.** A sagittal division separates right and left portions.
- 22 Frontal (coronal).** A frontal division separates front and back portions.
- 23 Transverse (axial).** A transverse division separates upper and lower portions.
- 24 Midsagittal (median).** A midsagittal plane is a sagittal plane at the midline.
- 25 Parasagittal.** A parasagittal plane is parallel to, but offset from, the median plane.
- 26 Forward.** The palms face anteriorly in anatomical position.
- 27 Prone.** Prone describes a face-down position.
- 28 No.** The terms still refer to that person's own sides.

References used to verify terminology

National Cancer Institute, SEER Training Modules: Anatomical Terminology.
training.seer.cancer.gov/anatomy/body/terminology.html

OpenStax, Anatomy and Physiology 2e, section 1.6: Anatomical Terminology.
openstax.org/books/anatomy-and-physiology-2e/pages/1-6-anatomical-terminology

Original wording, diagrams, and exercises. Reference material was checked on 3 October 2026.